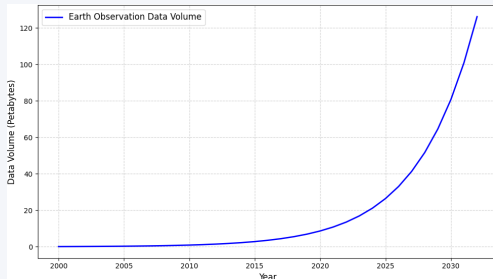


Management und FAIRe Bereitstellung von multi-dimensionalen analysefertigen Rasterdaten mittels dynamischer Webdienste

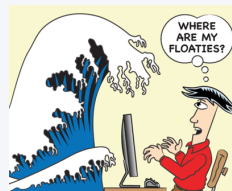
Markus Möller

Julius Kühn-Institut · Digitalisierung und Künstliche Intelligenz · AG Forschungsdatenmanagement · Kleinmachnow

Exponentieller Anstieg des EO-Datenvolumens



©perplexity.ai



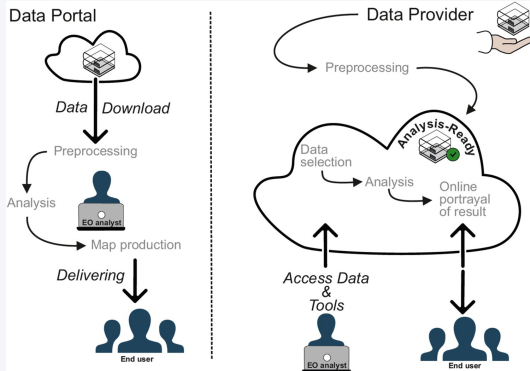
© S. Kearsley, 2012.
www.lowtechcartoon.com

Allein das Copernicus-Programm generiert täglich etwa 150 TB Rohdaten auf globaler Ebene ohne Berücksichtigung weiterverarbeiteter Daten.

Guo, H., 2017. Big Earth data: A new frontier in Earth and information sciences. Big Earth Data 1, 4–20.
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Apicella, L., De Martino, M., Quarati, A., 2022. Copernicus User Uptake: From Data to Applications. IJGI 11, 121. <https://doi.org/10.3390/ijgi11020121>

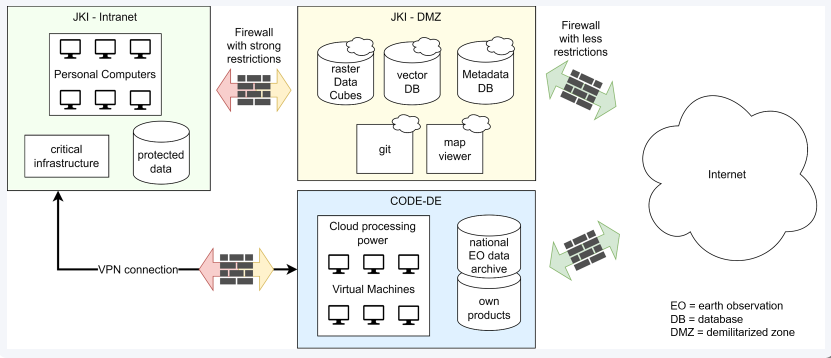
Paradigmenwechsel



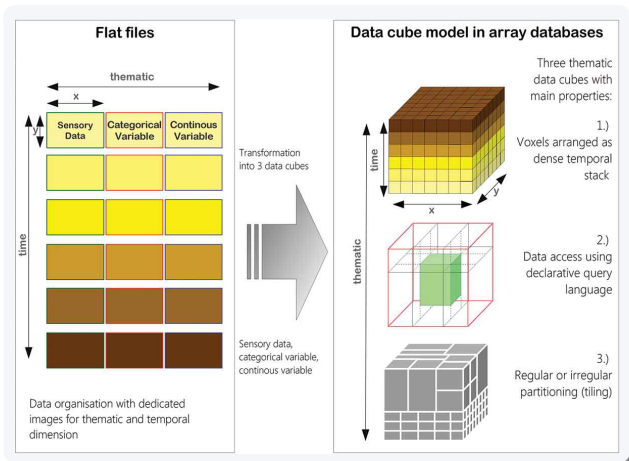
Sudmanns, M., Tiede, D.,
Lang, S., Bergstedt, H., Trost,
G., Augustin, H., Baraldi, A.,
Blaschke, T., 2020. Big Earth
data: disruptive changes in
Earth observation data
management and analysis?
International Journal of Digital
Earth 13, 832–850.

<https://doi.org/10.1080/17538947.2019.1585976>

Geodateninfrastruktur am JKI (JKI-GDI)



Beyer, F., Brandt, P., Schmidt, M., König, S., Stahl, U., Baumann, P., Golla, B., Gerighausen, H., Möller, M., 2025. Big geodata and spatial data infrastructures: A perspective of a German authority. under review

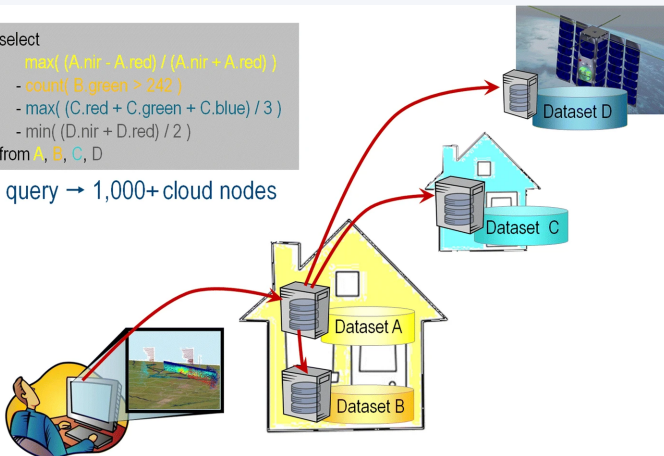


Tiede, D., Baraldi, A., Sudmanns, M., Belgiu, M., Lang, S., 2017. Architecture and prototypical implementation of a semantic querying system for big Earth observation image bases. *European Journal of Remote Sensing* 50, 452–463.
<https://doi.org/10.1080/22797254.2017.1357432>

JKI-GDI: JKI-Data Cube

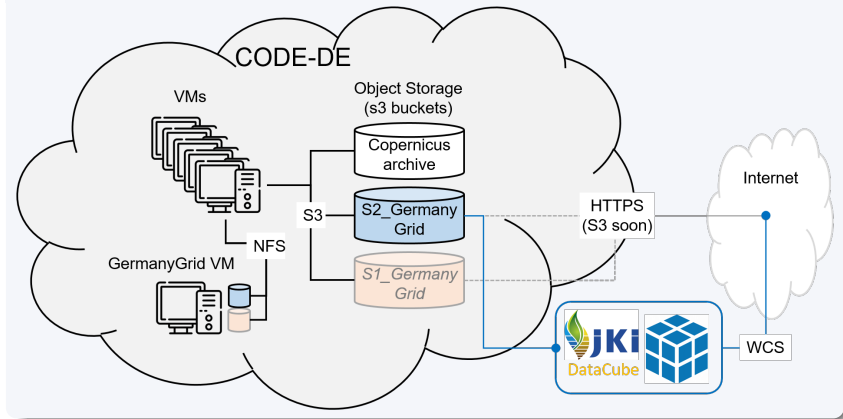
```
select  
  max( (A.nir - A.red) / (A.nir + A.red) )  
- count( B.green > 242 )  
- max( (C.red + C.green + C.blue) / 3 )  
- min( (D.nir + D.red) / 2 )  
from A, B, C, D
```

1 query → 1,000+ cloud nodes



Baumann, P., Escriu, J., 2019. INSPIRE coverages: an analysis and some suggestions. Open geospatial data, softw. stand. 4, 1. <https://doi.org/10.1186/s40965-019-0059-x>

CODE-DE und JKI Data Cube

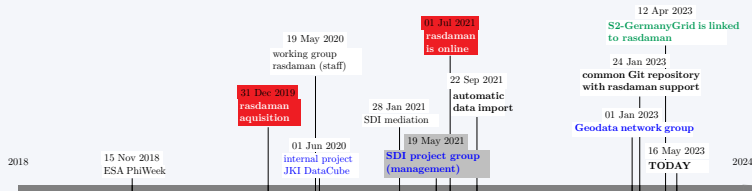


Beyer, F., Brandt, P., Schmidt, M., König, S., Stahl, U., Baumann, P., Golla, B., Gerighausen, H., Möller, M., 2025. Big geodata and spatial data infrastructures: A perspective of a German authority. under review

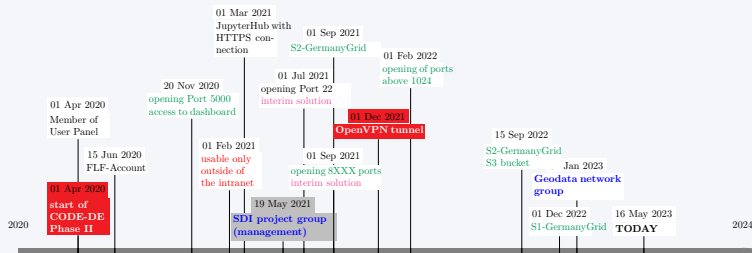
McClelland, J., Domenic, A.R., Beyer, F., de Kock, A., Golla, B., 2025. Developing a reusable infrastructure for machine learning on diverse earth observation data for sustainable agriculture and forestry. Lecture Notes in Informatics (LNI).https://gil-net.de/Publikationen/GIL25_Proceedings_final_01.pdf

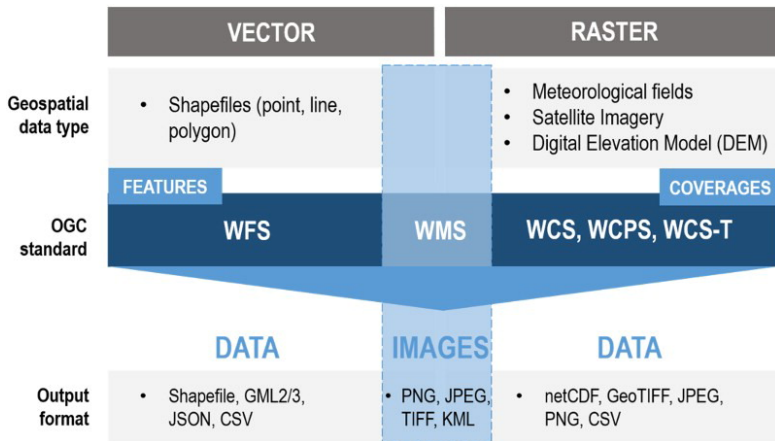
Big Geodata am JKI: Implementierung

JKI Data Cube



CODE-DE

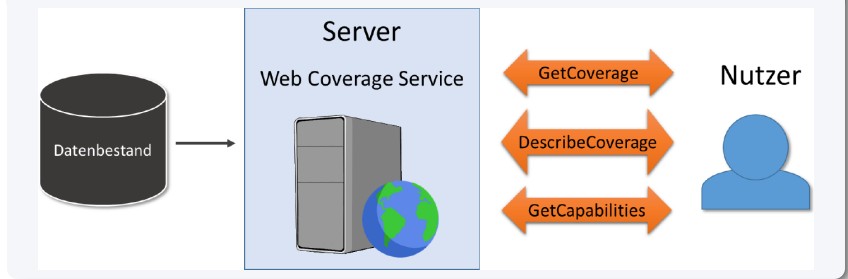




Wagemann, J., Clements, O., Marco Figuera, R., Rossi, A.P., Mantovani, S., 2018. Geospatial web services pave new ways for server-based on-demand access and processing of Big Earth Data. International Journal of Digital Earth 11, 7–25. <https://doi.org/10.1080/17538947.2017.1351583>

Baumann, P., 2021. A general conceptual framework for multi-dimensional spatio-temporal data sets. Environmental Modelling Software 143, 105096. <https://doi.org/10.1016/j.envsoft.2021.105096>

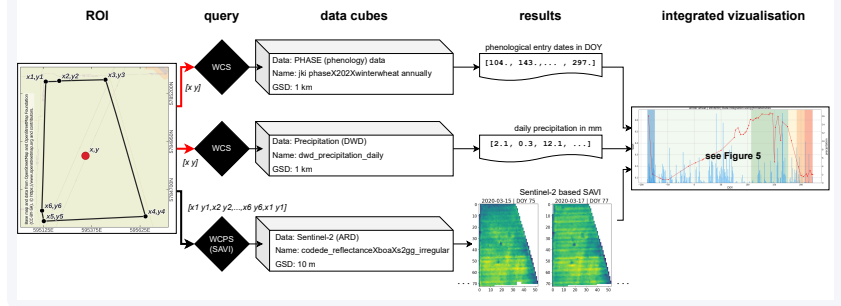
Web Coverage Service



The **Web Coverage Service (WCS)** is an interface standardised by the **Open Geospatial Consortium (OGC)** that enables access to spatial 'coverages' or multi-dimensional raster data.

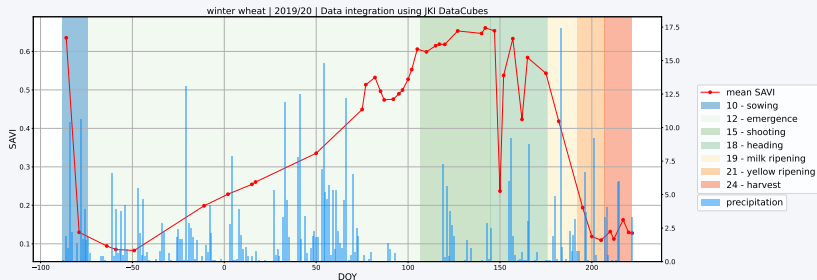
- **GetCapabilities:** provides an XML description of the service properties and data sets offered by the requested server,
- **DescribeCoverage:** provides XML descriptions of the coverages,
- **GetCoverage:** returns the requested coverage.

Standardisierte Webdienste für multi-dimensionale Rasterdaten



Beyer, F., Brandt, P., Schmidt, M., König, S., Stahl, U., Baumann, P., Golla, B., Gerighausen, H., Möller, M., 2025. Big geodata and spatial data infrastructures: A perspective of a German authority. under review

Standardisierte Webdienste für multi-dimensionale Rasterdaten



Beyer, F., Brandt, P., Schmidt, M., König, S., Stahl, U., Baumann, P., Golla, B., Gerighausen, H., Möller, M., 2025. Big geodata and spatial data infrastructures: A perspective of a German authority. under review

Beyer, F., Brandt, P., Schmidt, M., Stahl, U., Golla, B., Gerighausen, H., Möller, M., 2023. A paradigm shift towards decentralized cloud-integrated spatial data infrastructures: Lessons learned and solutions provided for public authorities. EarthArXiv. (preprint) <https://doi.org/https://doi.org/10.31223/X53H3N>

FAIRe Prozesse

Jupyter Notebook WC(P)S demo of JKI Data Cubes

Beyer, F., Möller, M., Dierks, M., 2024. JKIDataCubeDemo.
Zenodo. <https://doi.org/10.5281/zenodo.14215012>

FAIRe Daten

OpenAPI WCS documentation

- <https://sf.julius-kuehn.de/openapi/weather/>
- https://sf.julius-kuehn.de/openapi/S2_GermanyGrid_JKI/
- <https://sf.julius-kuehn.de/openapi/phase/>

Möller, M., Beyer, F., Dierks, M., Horney, P., Baumann, P., Svoboda, N., Gerstmann, H., 2023. Germany-wide time series of interpolated phenological observations of main crop types between 1993 and 2021. EarthArXiv. (preprint)
<https://doi.org/https://doi.org/10.31223/X5D37T>

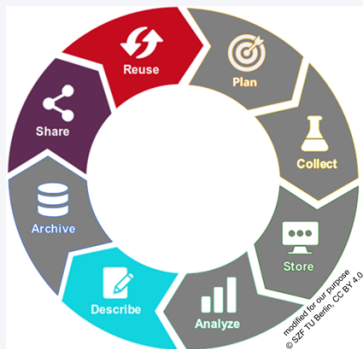
Beyer, F., Brandt, P., Schmidt, M., König, S., Stahl, U., Baumann, P., Golla, B., Gerighausen, H., Möller, M., 2025. Big geodata and spatial data infrastructures: A perspective of a German authority. under review

Datenlebenszyklus

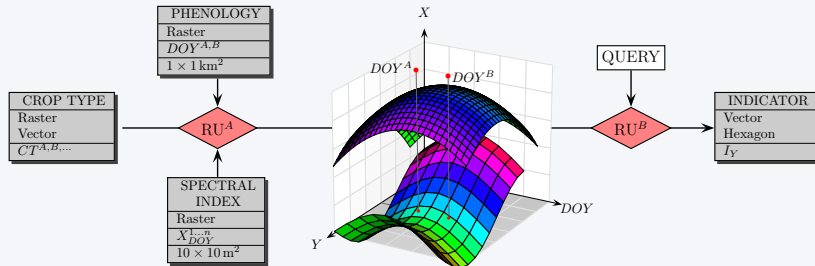


FAIRagro

<https://fairagro.net/en/publishing-research-code-fair-a-roadmap/>



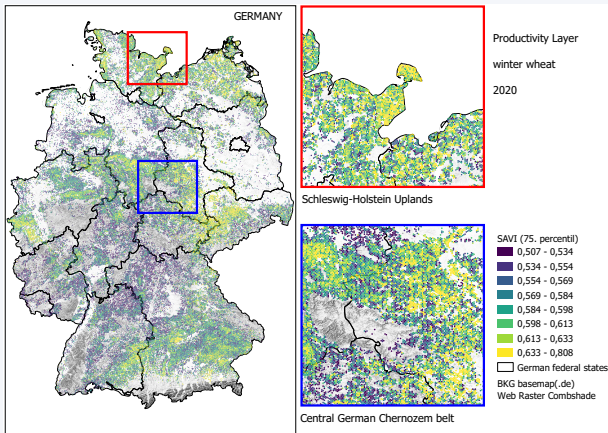
Biodiversitätsindikatoren: Produktivität



Categorie	Description	Data type	Resolution
PHENOLOGY	interpolated phenological entry DOYs of winter wheat phases <i>shooting</i> and <i>yellow ripeness</i>	raster	1 km
CROP TYPE	winter wheat in 2019, 2020 and 2021	raster	20 m
RU^A	winter wheat area within hexagon	raster	–
RU^B	hexagon	vector	1 km
SPECTRAL INDEX	<i>SAVI</i> based on S2_GermanyGrid	raster	20 m
QUERY	75% percentile of <i>SAVI</i> values within the vital development phase averaged over the hexagon	vector	1 km

Raum-zeitlich
dynamische
Parametrisierung

Biodiversitätsindikatoren: Produktivität



Klein, K. et al., 2024.
MonViA-Indikatorenbericht
2024: Bundesweites Monitoring
der biologischen Vielfalt in
Agrarlandschaften.
Verbundprojekt MonViA, Bonn.
[https://www.openagrar.de/
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Beyer, F., Brandt, P., Schmidt,
M., König, S., Stahl, U.,
Baumann, P., Golla, B.,
Gerighausen, H., Möller, M.,
2025. Big geodata and spatial
data infrastructures: A
perspective of a German
authority. under review

Erosionsmodellierung

Batista, P.V.G., Möller, M., Schmidt, K., Waldau, T., Seufferheld, K., Htitiou, A., Golla, B., Ebertseder, F., Auerswald, K., Fiener, P., 2025. Towards regional nowcasting of soil-erosion events on arable land. Research Square. CATENA, 109080. <https://doi.org/10.1016/j.catena.2025.109080>; Source code: <https://doi.org/10.5281/zenodo.11072308>

Ertragsabschätzung

Brandt, P., Beyer, F., Borrmann, P., Möller, M., Gerighausen, H., 2024. Ensemble learning-based crop yield estimation: a scalable approach for supporting agricultural statistics. GIScience Remote Sensing 61, 2367808. <https://doi.org/10.1080/15481603.2024.2367808>

Extremwetter und Ertragsverluste

Riedesel, L., Möller, M., Horney, P., Golla, B., Piepho, H.-P., Kautz, T., Feike, T., 2023. Timing and intensity of heat and drought stress determine wheat yield losses in Germany. PLoS ONE 18, e0288202. <https://doi.org/10.1371/journal.pone.0288202>

Riedesel, L., Möller, M., Piepho, H.-P., Rentel, D., Lichthardt, C., Golla, B., Kautz, T., Feike, T., 2024. Site conditions determine heat and drought induced yield losses in wheat and rye in Germany. Environ. Res. Lett. 19, 034024. <https://doi.org/10.1088/1748-9326/ad24d0>

Riedesel, L., Ma, D., Piepho, H.-P., Laidig, F., Möller, M., Golla, B., Kautz, T., Feike, T., 2024. Climate change induced heat and drought stress hamper climate change mitigation in German cereal production. Field Crops Research 317, 109551. <https://doi.org/10.1016/j.fcr.2024.109551>